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Module 2 Introduction
Lesson 1: Estimating Probabilities and Posterior Distributions
Lesson 2: Summarizing Posterior Inference and Prior Distributions
Lesson 3: Normal Distribution and Other Single-Parameter Models
Reading: Normal Distribution and Other Single-Parameter Models Reading 2h
Video: Other Single-Parameter Models Pt. 1 2 min
Video: Other Single-Parameter Models Pt. 2 9 min
Video: Other Single-Parameter Models Pt. 3 7 min
Video: Other Single-Parameter Models Pt. 4 9 min
Video: Other Single-Parameter Models Pt. 5 4 min
Video: Other Single-Parameter Models Pt. 6 9 min
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Practice Assignment: Normal Distribution and Other Single-Parameter Models Quiz Submitted
Lab: Module 2 - Lesson 3 - RStudio Lab 1h
Module 2 Summative Assessment
Module 2 Summary

Your grade: 100%

Your latest: 100% • Your highest score: 100% • Let's review all of your answers. Next item

Normal Distribution and Other Single-Parameter Models Quiz

Instructions

Review Learning Objectives

1. Suppose $y|\theta$ is exponentially distributed with rate θ, and the prior distribution of θ is $Gamma\,(\alpha, \beta)$. Suppose we know our observations, $\mathbf{y}$, is larger than 200 but we do not know the exact amount. What is the posterior distribution? (Hint: Consider all possible amounts of y larger than 200.) **1 / 1 point**

Ready to review what you've learned before starting the assignment? I'm here to help.

Help me practice Let's chat

- (selected) $Gamma\,(\alpha, \beta + 200)$
- $Gamma\,(\alpha, \beta)$
- It is not possible to compute.
- $Gamma\,(\alpha + 1, \beta + 200)$

Correct
Correct!

Assignment details

Submitted Nov 21, 12:53 AM IST
Attempts Unlimited
Retry

Your grade
To pass you need at least 80%. We keep your highest score.
View submission
100%

2. Suppose you are working with a problem whose likelihood follows a normal with unknown mean μ and known variance σ^2. Furthermore, suppose you use an noninformative prior distribution (defined finitely on the number line). What kind of distribution is the posterior distribution? See feedback **1 / 1 point**

- Beta
- (selected) None of the above
- Exponential
- Gamma

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Correct
Correct! Because the uniform distribution is defined finitely on the number line, it is not conjugate. Multiplied by the normal distribution, we end up with something truncated, which is different entirely.

3. Precision is the inverse of the standard deviation. **1 / 1 point**

- (selected) False
- True

Correct
Correct! Precision is the inverse of the variance.